Re-rating a $15B book

Track A

Excel-lence

DNA India Decode 2026

Executive Summary

Re-rating every account to its own risk earns $2.6B of contribution, $1.5B more than doing nothing, and clears the 85% MLR ceiling with no accounts cut deliberately.

- **The book looks healthy, but its price no longer tracks risk.** A $15B, 2.2 million life book, diversified across every industry, region and account size, sits at 88.7% MLR, only 3.7 points over its ceiling. Yet premium per life is almost flat, 1.3x across the book, while claims climb 16x with risk, so the healthy are overcharged and the high risk run underwater.

- **We reprice each account off a single lever, the renewal quote.** Maximising book contribution hands every account one rate that covers its claims and marks up how captive it is, while staying inside the 80% retention floor and the 85% MLR ceiling. The one input we estimate, how renewal responds to price, is calibrated and causal.

- **The book earns $2.6bn, worth $1.5bn more than doing nothing.** Against a flat, do-nothing quote that erodes the book to $1.1B and lets MLR drift to 91.9%, our per account rate earns $2.6B of contribution, brings book MLR from 88.7% down to 80.9%, holds retention at 80.6%, and cuts no account deliberately. It is the only strategy we tested that lands below the 85% ceiling without shedding the book.

- **The result holds under stress.** It is stable across 5,000 simulated renewal seasons, and under a completely different, tree-based renewal model it keeps 99% of its value ($2,603M versus $2,624M), with 99% of accounts moving by at most one rate step.

A book that looks healthy on every conventional lens

Diversified across industry, geography, account size and renewal timing

The Client

- Large-group employer medical insurer
  - Licensed 2007, Columbus OH
  - ~9,500 staff
- Sold through brokers, priced on **experience**

2.2M lives

7,300 accounts

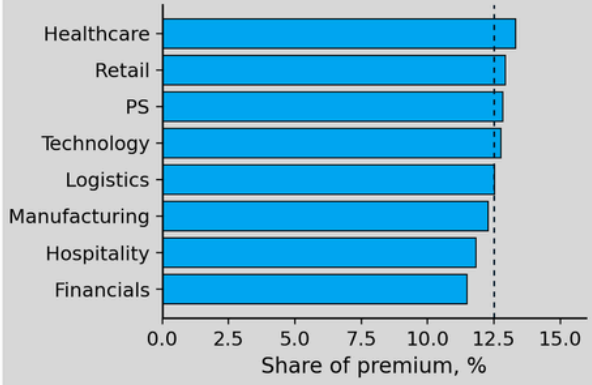
$15B premiums

48 regions

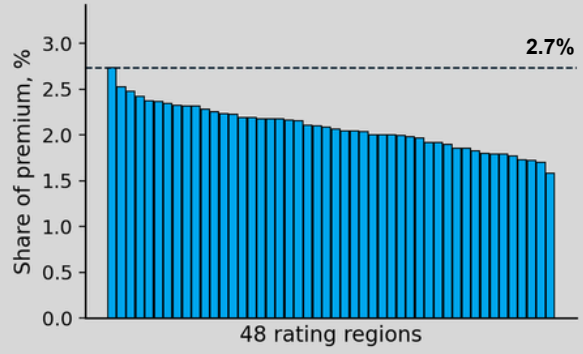
- Since 2015, underpriced incumbents on new-business quotes to win share. It bought volume, not margin
- The book runs **88.7% MLR**, just 3.7 points over the 85%. Small enough that a flat **~4.4% looks like the fix**
- **But...**
  The problem isn't the book; it's hiding between accounts, **in the pricing**

No concentration anywhere: the book is evenly spread on every cut

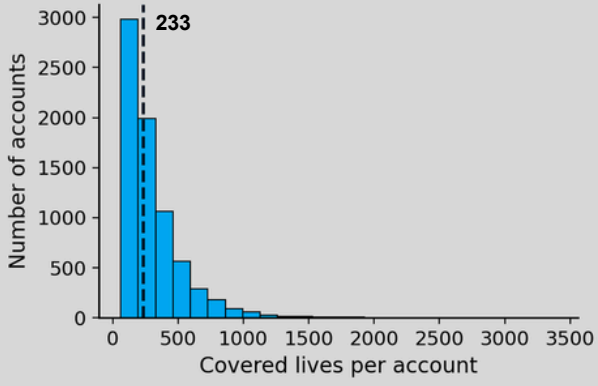
Premium spread evenly across 8 industries



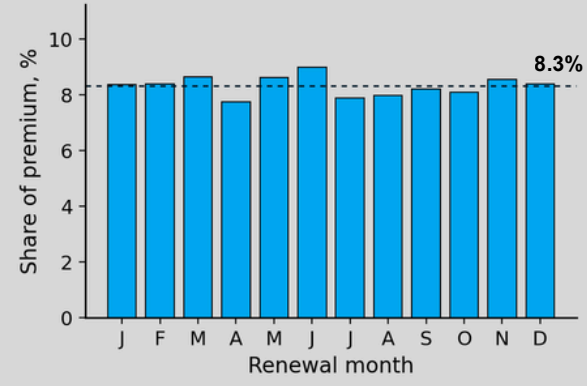
No region over 2.7%, fragmented across 48



Thousands of small accounts; median at 233



Renewals steady all year; no month >9%



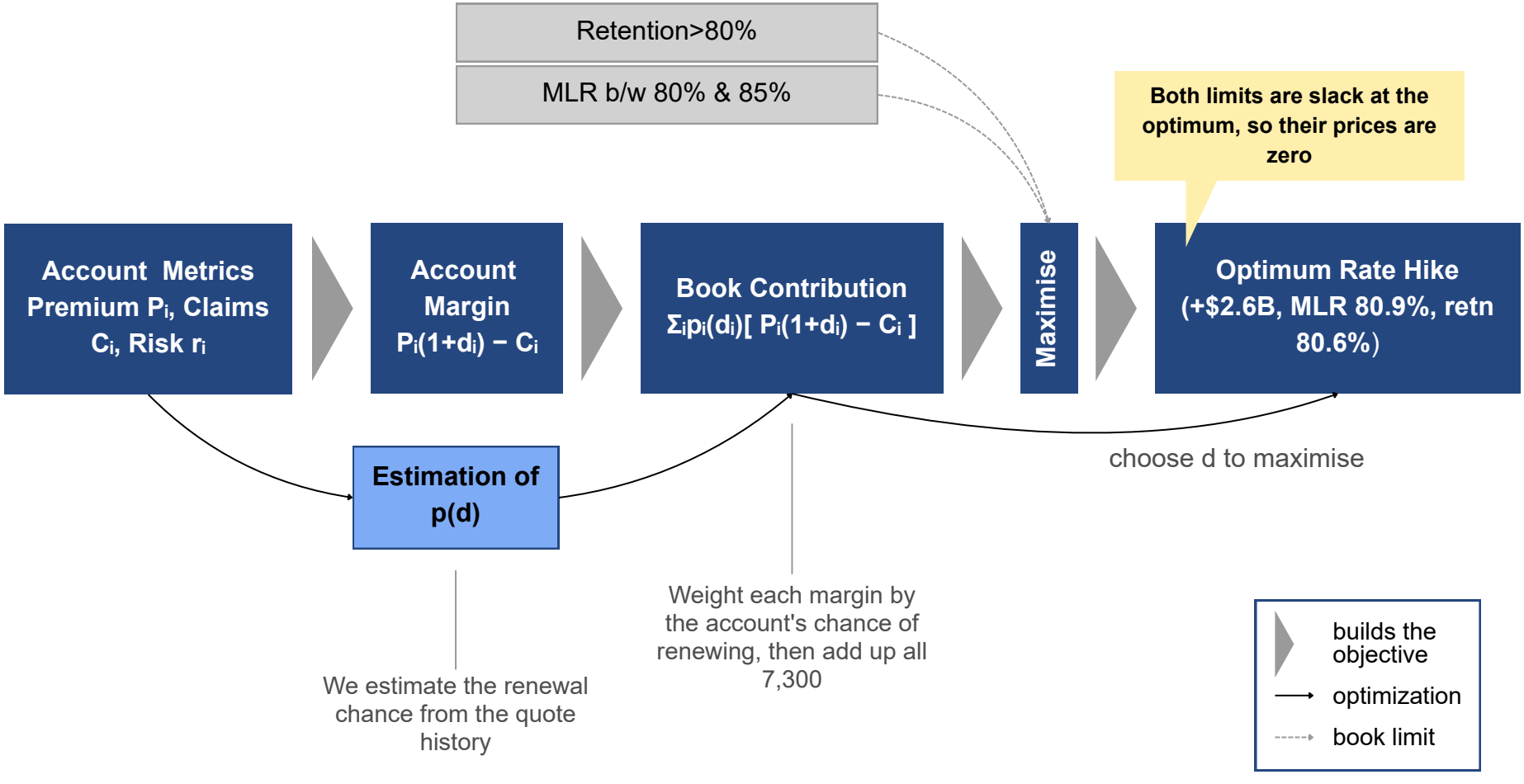
But underneath, price barely tracks risk: the healthy overpay and the high-risk run underwater

Bandwise P/L ($ M) 5,592 -1,208 -2,690 = 1694



We rate each account to maximise its expected contribution

The engine end to end: from each account's premium, claims and risk to its optimal rate

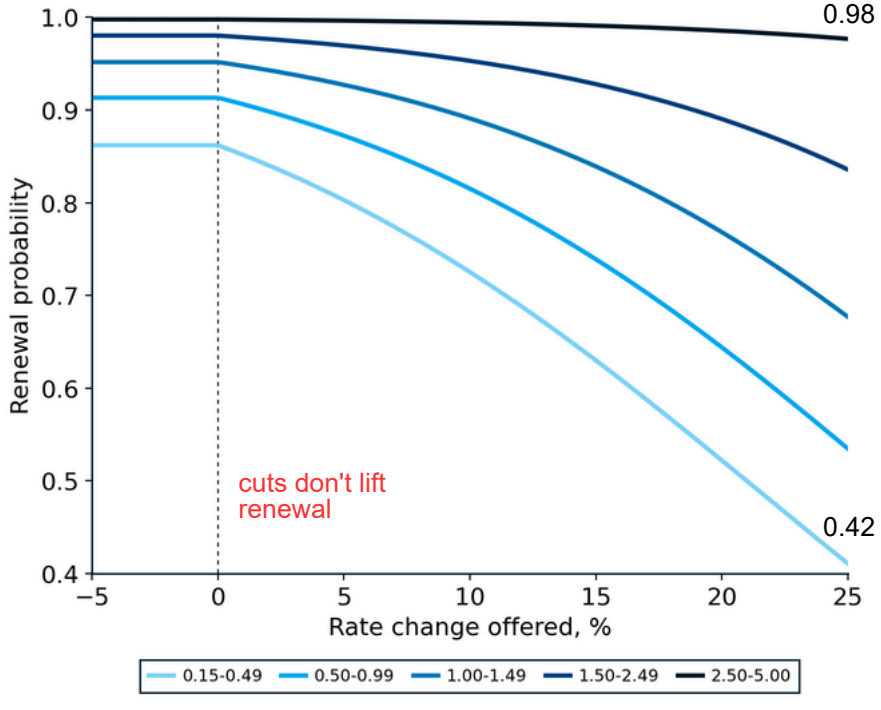


The one input we estimate is how renewal responds to price

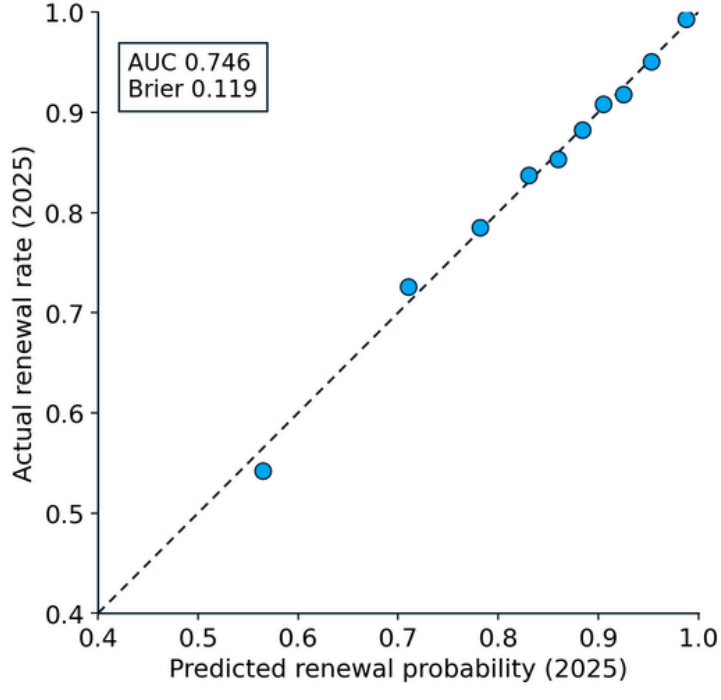
Renewal probability by risk band, and it holds well when tested out of sample

$$p(d) = 1 / (1 + e^{-z}) \qquad z = \beta_0 + \beta_1 \cdot d^+ + \beta_2 \cdot (d^+)^2 + \beta_3 \cdot d^+ \cdot r + \beta_4 \cdot r,\ d^+ = \max(d, 0)$$

The healthy are elastic; the sickest renew regardless



Predicted and actual renewal line up on 2025 (OOS)



Each account's rate is three forces: cover its claims, mark up how captive it is, rebate what the book needs

$$\Pi = \Sigma_i\, p_i(d_i) \cdot [\, P_i(1 + d_i) - C_i \,]$$

chance it renews at rate d_i

margin kept if it renews

The book-level condition separates into one equation per account

MAXIMISE ($\partial\Pi/\partial d_i = 0$)

$$d^*_i = (\ell_i - 1) + 1/\eta_i - (\lambda, \mu)/P_i$$

clipped to [−15%, +25%]

$-p'/p$

shadow prices of the floor / ceiling

Break-even $(\ell_i - 1)$

- **Cover** the account's **own claims**. Set by loss ratio ℓ = C/P
- Large on the sick, negative on the already-profitable healthy

Elasticity $1/\eta_i$

- The **extra margin** renewal actually allows
- Wide for captive accounts, thin for shoppable ones
- The only force that lifts a healthy account

Book Limits $(\lambda, \mu)/P$

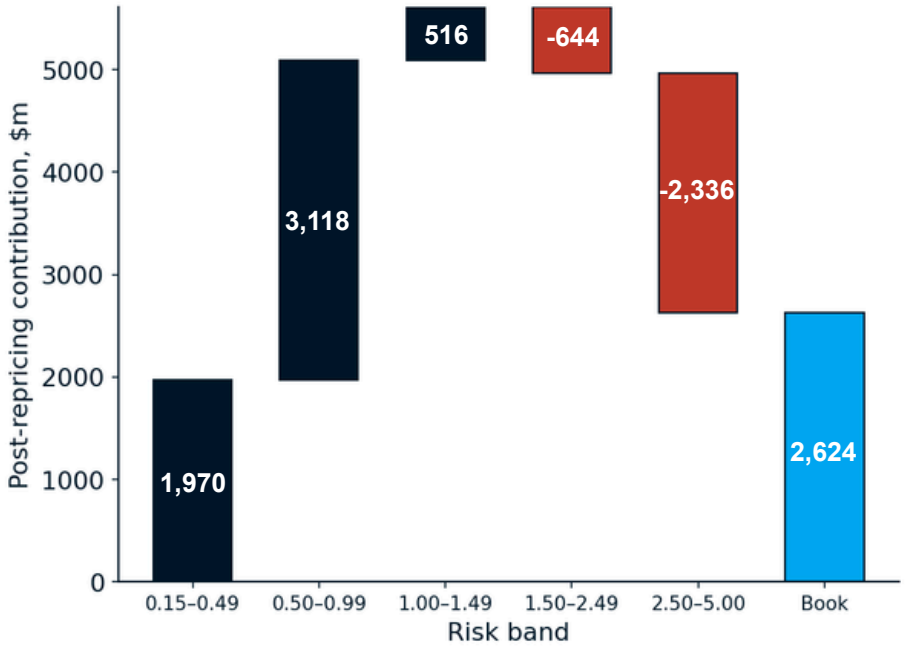
- Protects the 80% retention floor and 85% MLR ceiling
- **Both slack at our optimum** → zero for every account

Note: Find the complete optimization math here

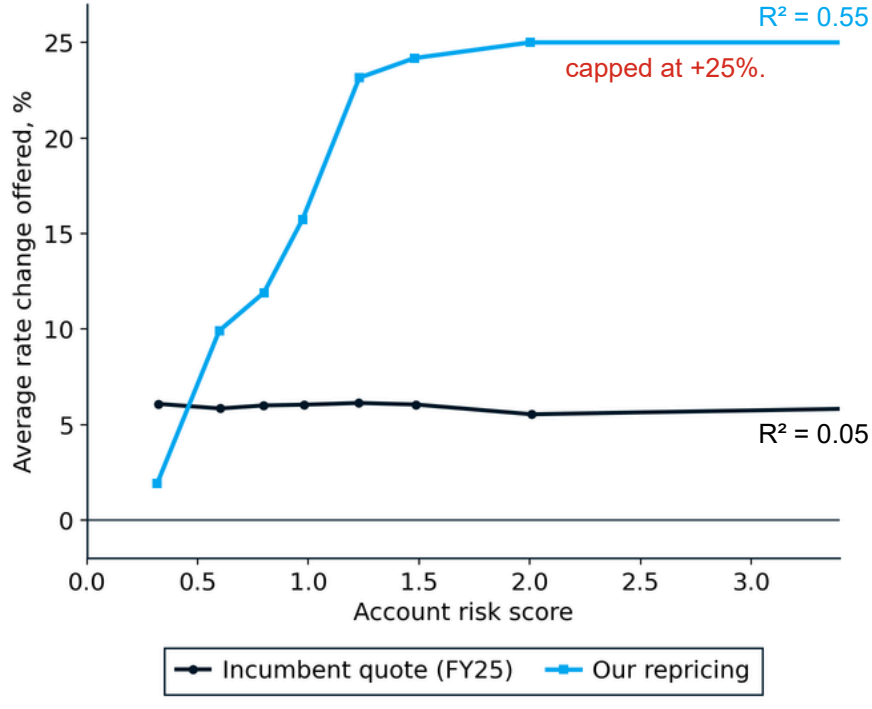
Repricing turns the book into $2.6B of contribution

Where the dollars come from across risk bands, and the pricing behind them

Healthy bands fund it; the sick tail gives back $3.0B



Where the incumbent held flat, our rate follows risk

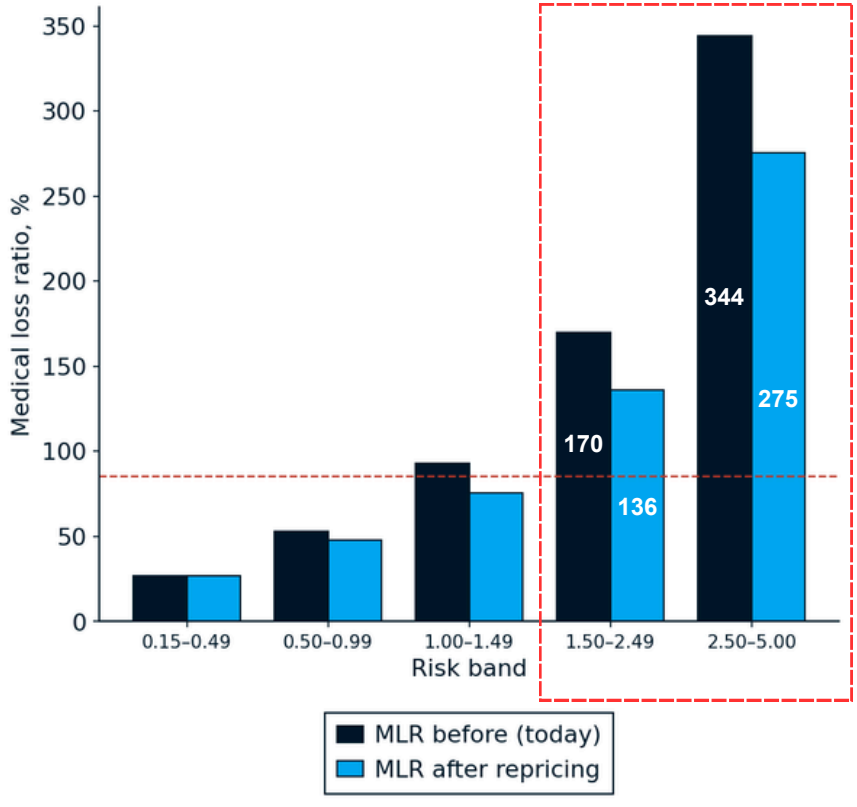


Pricing on experience adds about $0.9bn vs today's book and brings the MLR to 80.9%, below the 85% ceiling

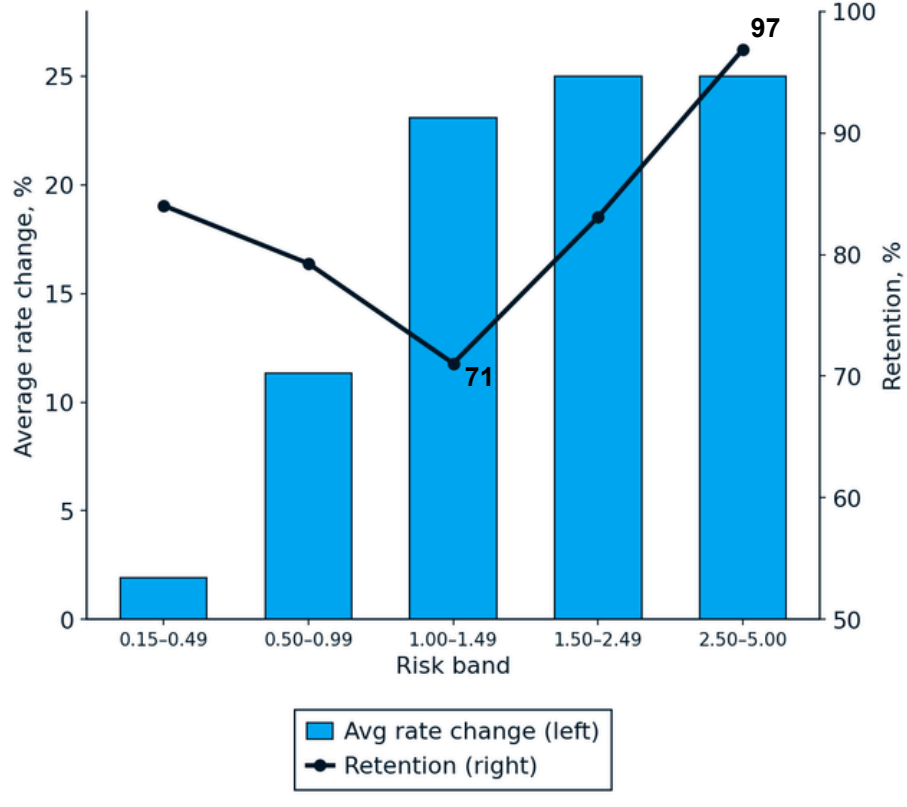
Price finally tracks risk across the book

MLR before and after repricing, and how retention responds by band

Every band's MLR falls; the two riskiest stay above 100% at the cap

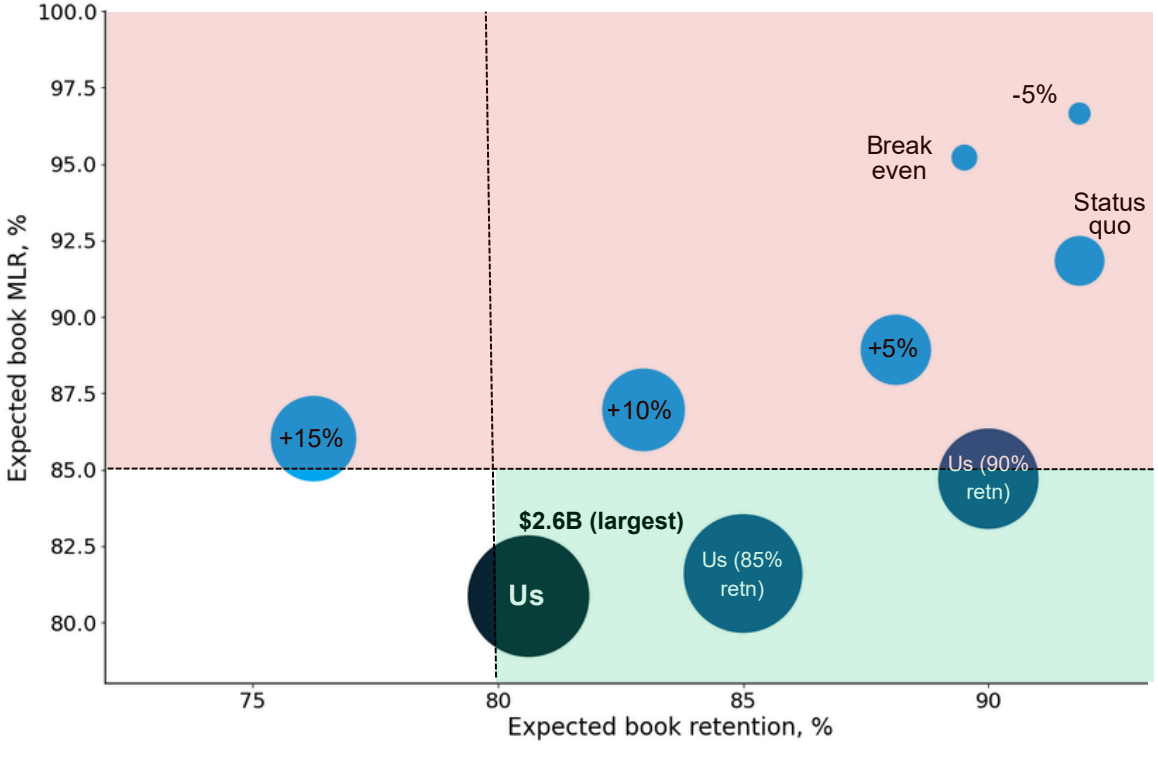


The full +25% lands on the captive bands, and they still renew



Among the strategies we tested, per-account repricing clears the ceiling without deliberately shedding the book

All of our variants sit in the safe zone below 85% MLR and above 80% retention



*Bubble sizes indicate book contribution

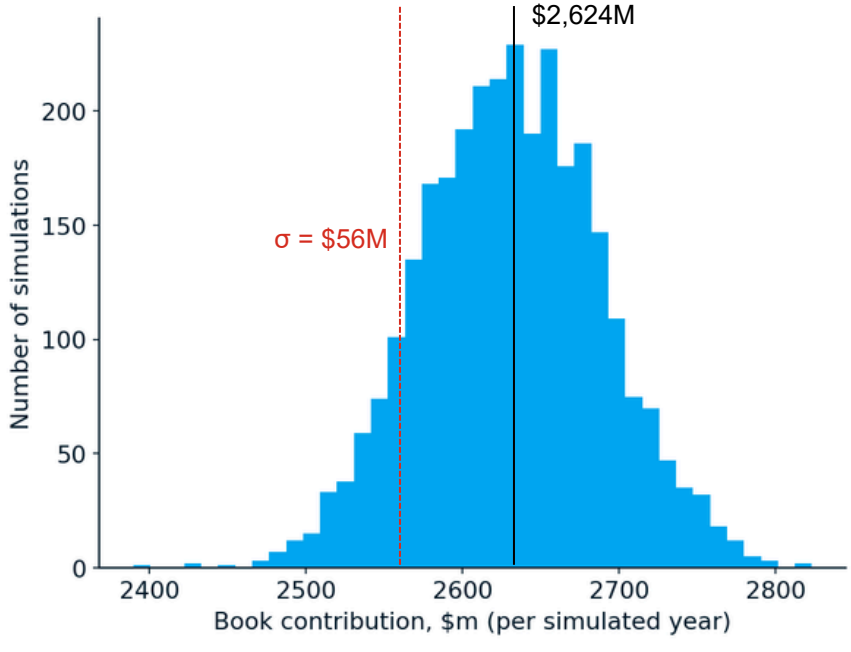
Why others lose?

- Raising everyone the same **overcharges the healthy accounts that can shop**, so retention falls faster than MLR improves. Even a flat +15% still sits above the ceiling at 86% MLR and sheds retention to 76%.
- **Cutting price buys nothing.** Renewal responds only to increases, so a 5% cut holds the exact same accounts as standing still.

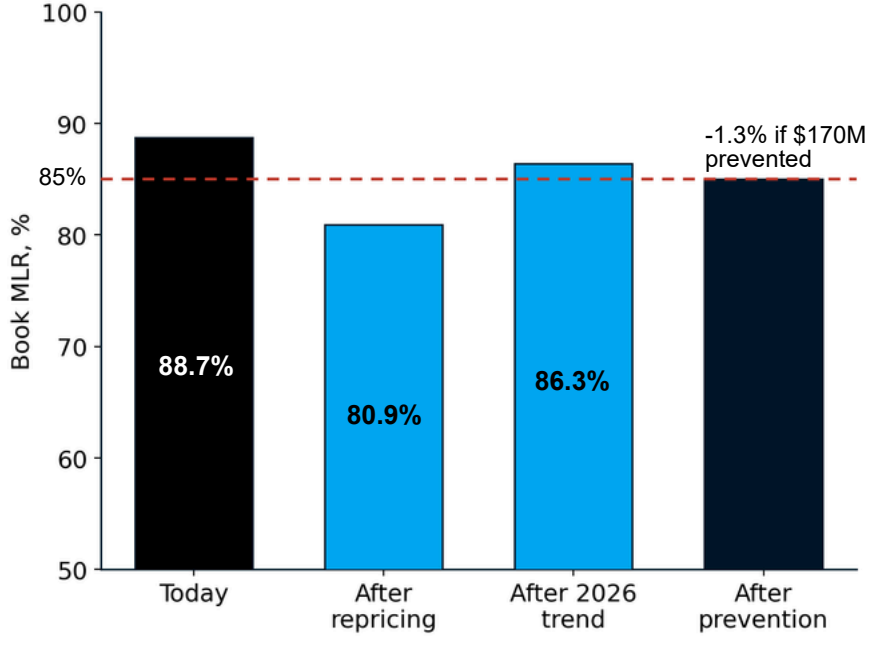
The answer holds: stable under simulation

5,000 simulated renewal seasons, and the book MLR path under 2026 medical trend

90% of seasons within ~$187M; the worst season still clears +$2.4B



Inflation lifts MLR to 86.3%, and $170M of prevented claims restores the ceiling



Thank You

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